Introduction to Supervised Machine Learning

QLS–MiCM Workshop

Instructor: [Your Name]

Duration: 4 hours total

Section: Introduction (~15 minutes)

Workshop Overview

1. Introduction & Motivation
2. Linear Regression
3. Regularization
4. Feature Engineering & Nonlinearity
5. MLPs for Tabular Data
6. Training Mechanics & Evaluation
7. Mini Hackathon

Why Machine Learning in Medicine?

- We generate **massive amounts of biomedical data**
  - Clinical measures, omics, imaging, EHRs
- ML can:
  - **Predict** outcomes (e.g., risk, response)
  - **Identify** patterns and biomarkers
  - **Support** clinical decisions

What Is Supervised Learning?

We have:

- **Features (X)** -> age, BMI, gene expression
- **Labels (y)** -> blood pressure, disease status

Goal:

> Learn a function that maps X -> y
> and generalizes to unseen data

Two Core Tasks

Task	Example	Output
Regression	Predict blood pressure	Continuous
Classification	Predict disease yes/no	Categorical

The Supervised ML Workflow

Data -> Preprocess -> Split -> Train -> Evaluate -> Interpret

- Clean data
- Train/test split
- Fit model
- Evaluate on unseen data
- Interpret results

Data–Model–Evaluation Triangle

```
 Data
/   \
```

Model ----- Evaluation

- Data quality defines limits
- Model choice shapes behavior
- Evaluation builds trust

Responsible ML in Biomedicine

- Watch for **bias** in datasets
- Maintain **privacy and consent**
- Strive for **interpretability**

Ethics is part of technical quality.

Tools for This Workshop

- **Python + Jupyter Notebooks**
- **scikit-learn**, **pandas**, **matplotlib**
- Example datasets: `Exercises/data/`
- Hands-on exercises: `Exercises/notebooks/`

By the End of Today

You'll be able to:

- Explain supervised learning basics
- Build simple regression & classification models
- Understand regularization & nonlinearity
- Evaluate and interpret model performance

Let's Get Started 🚀

Open your notebook:

`00_intro_and_data_exploration.ipynb`

We'll explore our first dataset together!